in technology have helped to move from magnetic tapes to hard disk drives to NVMe-based SSDs. The storage elements have moved from magnetic devices/ferroelectrics to semiconductor elements, and the new one being flash memory devices with multi-level memory devices.

Advantages of NVMe

NVMe is the prominent interface to the high-speed storage. PCIe interface that supports NVMe has become the default choice for building the SSD storage. This module consisting of PCIe bus, NVMe controller, and flash storage memories has distinguished advantages over many of the other storage systems.

The latency challenges are being optimised at various stages of HPC systems through emerging standards like CXL that include on the chip, chip to chip, boards to boards, and rack to rack within and across data centres.

To increase the throughput, the most prominent across the interfaces of many applications is PCIe, which is coming up with higher data rates for a given channel, and the latest upcoming specification will target 64Gbps per lane. This will help increase the throughput at the host site and the communication between the host and chip with a native controller on the chip. The PCIe, which acts as a bridge between the CPU and memory through the NVMe controller, has been the natural choice for building the future converged systems with integrated storage solutions.

Having discussed the main features and the storage solutions that are emerging, we can address the demands of large storage systems comprising the SSD storage device.

The emerging systems and the complex applications draw the attention of specific features of the SSD-based storage solutions for further reducing the latency in applications like ML, addressing the security at hardware and software levels, reducing the throughput requirements between CPU and storage, distributed storage over the fabric, semi-processed data near storage, etc.

Computing	Networking	Storage
Many cores	Networking intra-chip	High throughput
Core to core communication	Inter-chip (NOC)	Low latency
Native PCIe	NIC (blade to blade or server to server)	Fast read/write cycles
Memory controller	CXL interface for low latency NIC (Financial trading)	Low power, no moving parts
DRAM controller	MIPI, USB	Security hardware/software
CXL	Read only or write only option through controller	Data processing closer to storage
—	—	No encryption needed as it is close proximity device in case of transferjet

Table 1: The essentials of hyperconvergence systems/infrastructure

In other words, moving the data away from computation helps in saving the bandwidth, power, processing the metadata, security, and reducing latency.

Having a camera interface: Interfacing camera controllers to the storage device and transferring/analysing the data through a PCIe bus that seamlessly connects within the accelerators for graphics/images enhances the system performance by reducing the workload at the main processors and eases the data transfer between storage and computing.

Also, the recent emergence of U.2, M.2, and EDSFF form factors further enhance the potential usage of NVMe-based SSD devices for all applications.

The table above gives the different elements and options that can form the enterprise computing systems of the future.

A careful look at the constituents of the future high-performance computing (HPC) systems infers that the thin line between the compute to networking, networking to storage, and computing to storage is diminishing.

In reducing the latency across the ingredients of the hyperconverged systems, the NVMe-based flash storage systems play a critical role and emerge as the choice for all the storage solutions. The inherent advantages of the PCIe interface have become the de facto choice in the hyperconverged system.

To add to this, the NVMe controller has the adaptability to SSD architecture with the latest interface, memory controllers, advances in flash cells, and native controllers in CPUs. On top of all these, the controller is critical for customising for the umpteen number of applications that can be envisaged.

With the tsunami of data in every application, a highly efficient data storage drive is going to be the de facto choice for all the future gadgets and technologies like mobiles, laptops, HPC systems, edge computing, AI, ML, single board computing, car infotainment, and so on. END

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